

STRUCTURED ENQUIRY

Problem Statement:

Design a Contactless Liquid Level Controller using the LPC2148 that monitors the height of water in a tank using a suitable sensor and automatically controls the Water Inlet Valve to prevent overflow and dry-running conditions.

Team

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INTRODUCTION:

The Contactless Liquid Level Controller is an intelligent water management system developed using the LPC2148 microcontroller. The system uses an Ultrasonic Sensor to measure the water level in a tank without direct contact with the liquid. Based on the measured level, the LPC2148 automatically controls a DC Motor (Water Pump) to maintain the required water level.

The controller starts the motor when the water level drops below the minimum limit and stops it when the tank becomes full. This automation helps save water, reduces manual monitoring, and improves the efficiency of water storage system

OBJECTIVE:

1. Monitor the water level continuously.
2. Measure liquid level without physical contact.
3. Automatically operate the DC Motor for tank filling.
4. Avoid water overflow conditions.
5. Protect the pump from dry-running.
6. Display water level and motor status on LCD.
7. Improve water utilization and system reliability.

COMPONENTS USED:

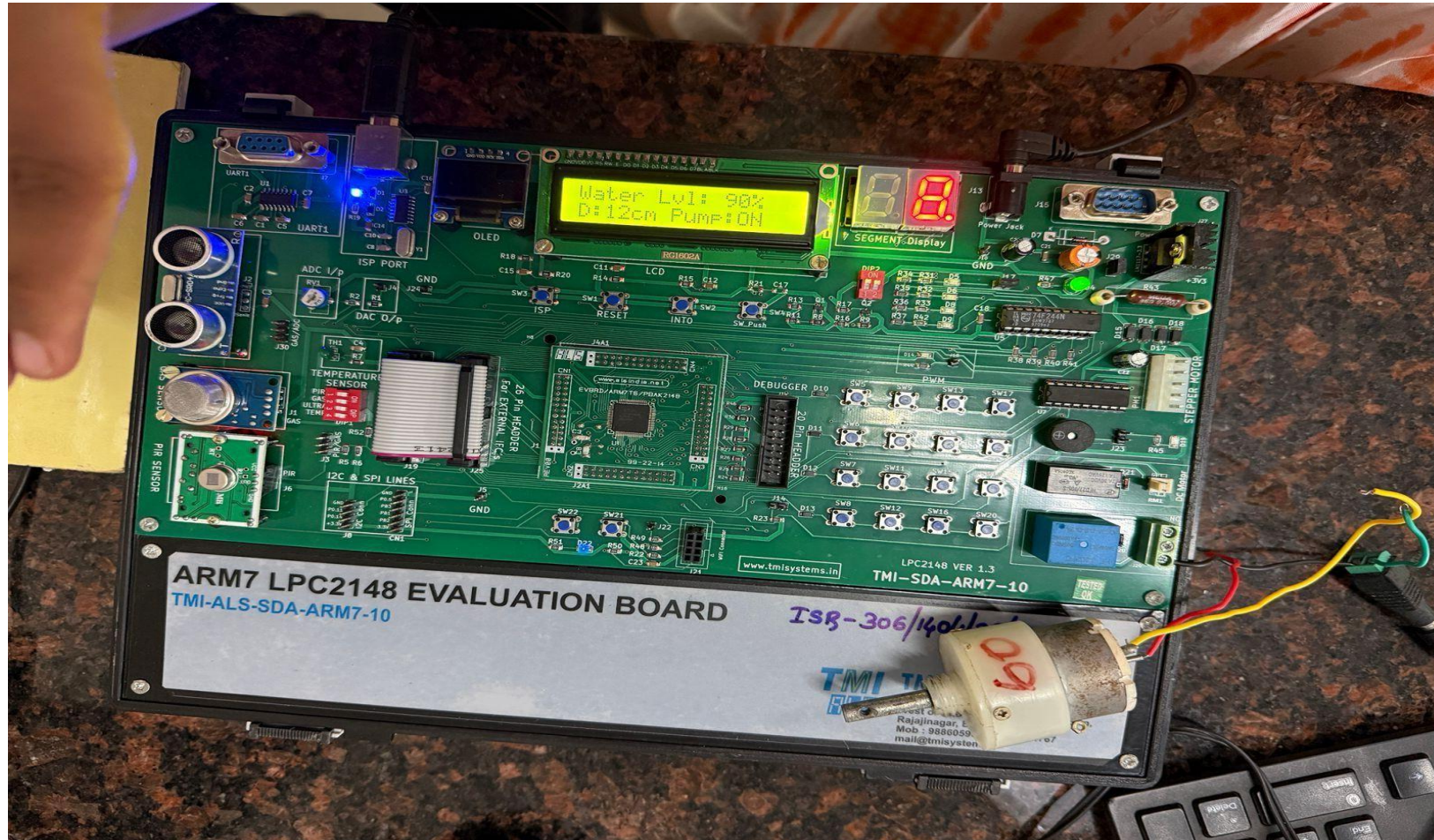
1. LPC2148 Microcontroller
2. Ultrasonic Sensor (HC-SR04)
3. DC Motor (Water Pump)
4. L293D Motor Driver IC
5. 16×2 LCD Display
6. Power Supply Circuit

WORKING PRINCIPLE:

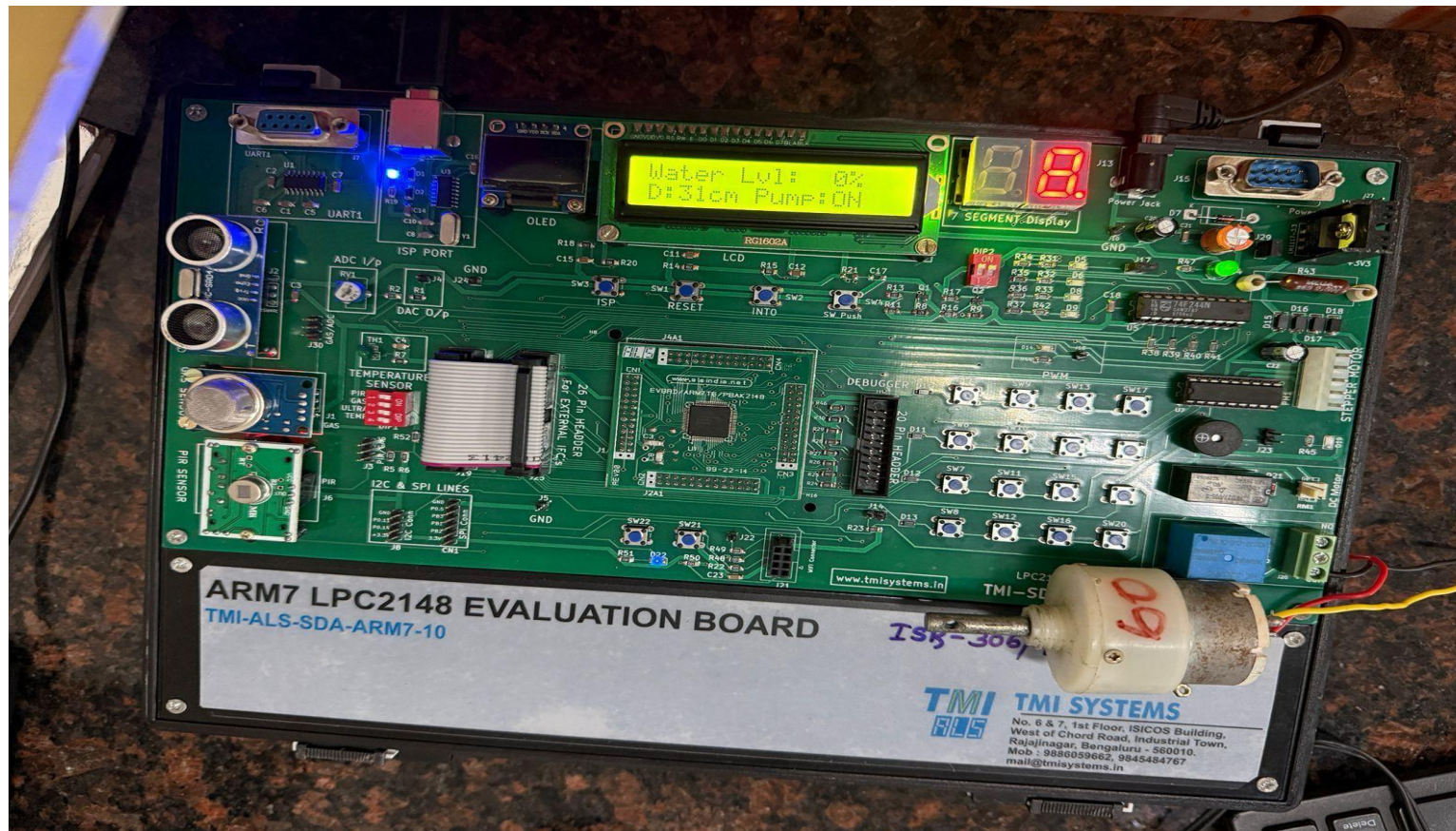
1. The Ultrasonic Sensor is fixed at the top of the water tank.
2. It continuously measures the distance between the sensor and the water surface.
3. The LPC2148 calculates the current water level from the measured distance.
4. If the water level falls below the preset minimum level, the controller switches ON the DC Motor.

- 5.The motor pumps water into the tank until the maximum level is reached.
6. Once the tank is full, the controller switches OFF the motor automatically.
- 7.The LCD continuously displays the water level and motor condition.

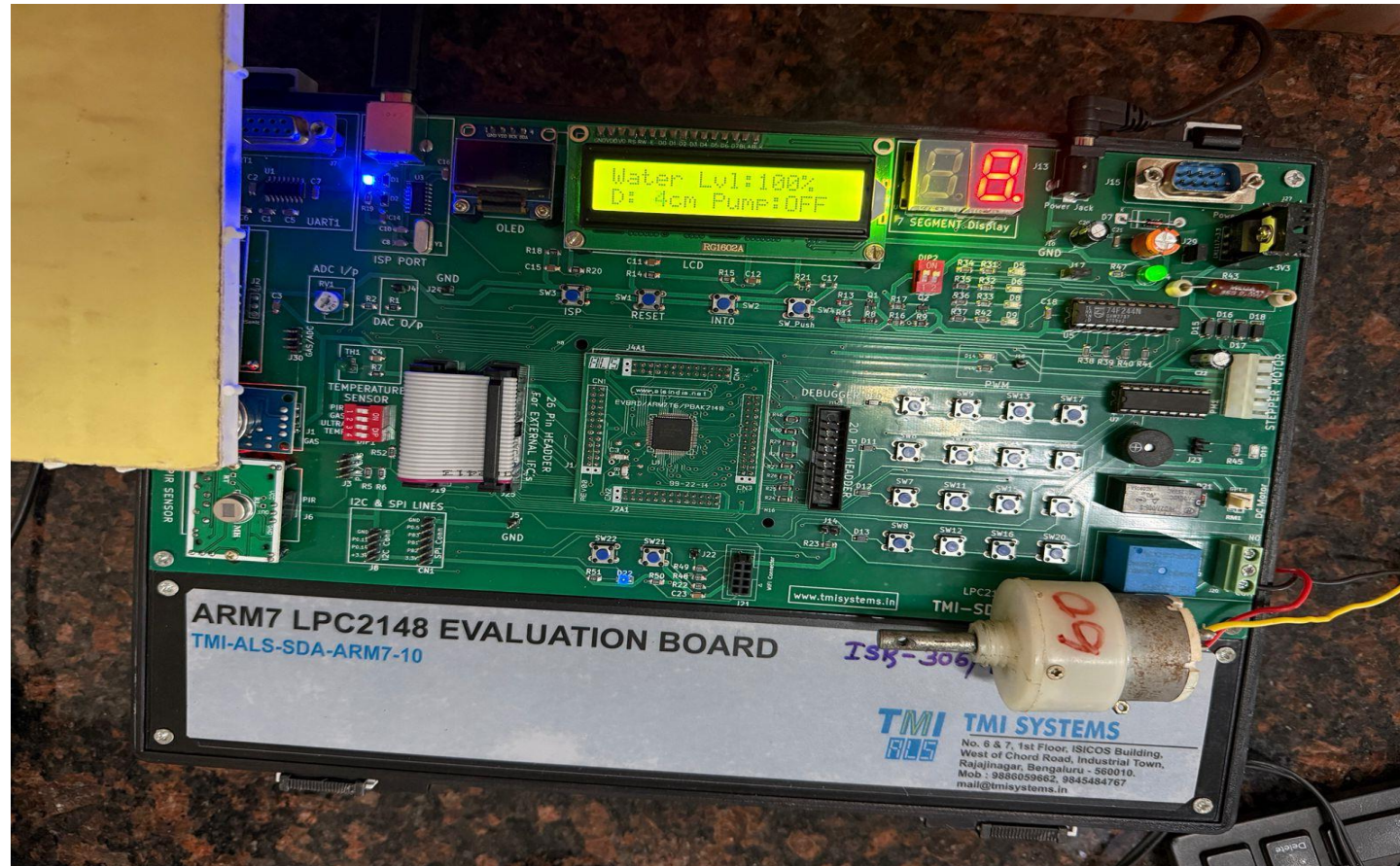
- When the water level is 90%
- The valve is on



- When the water level is 0%
- The valve is on

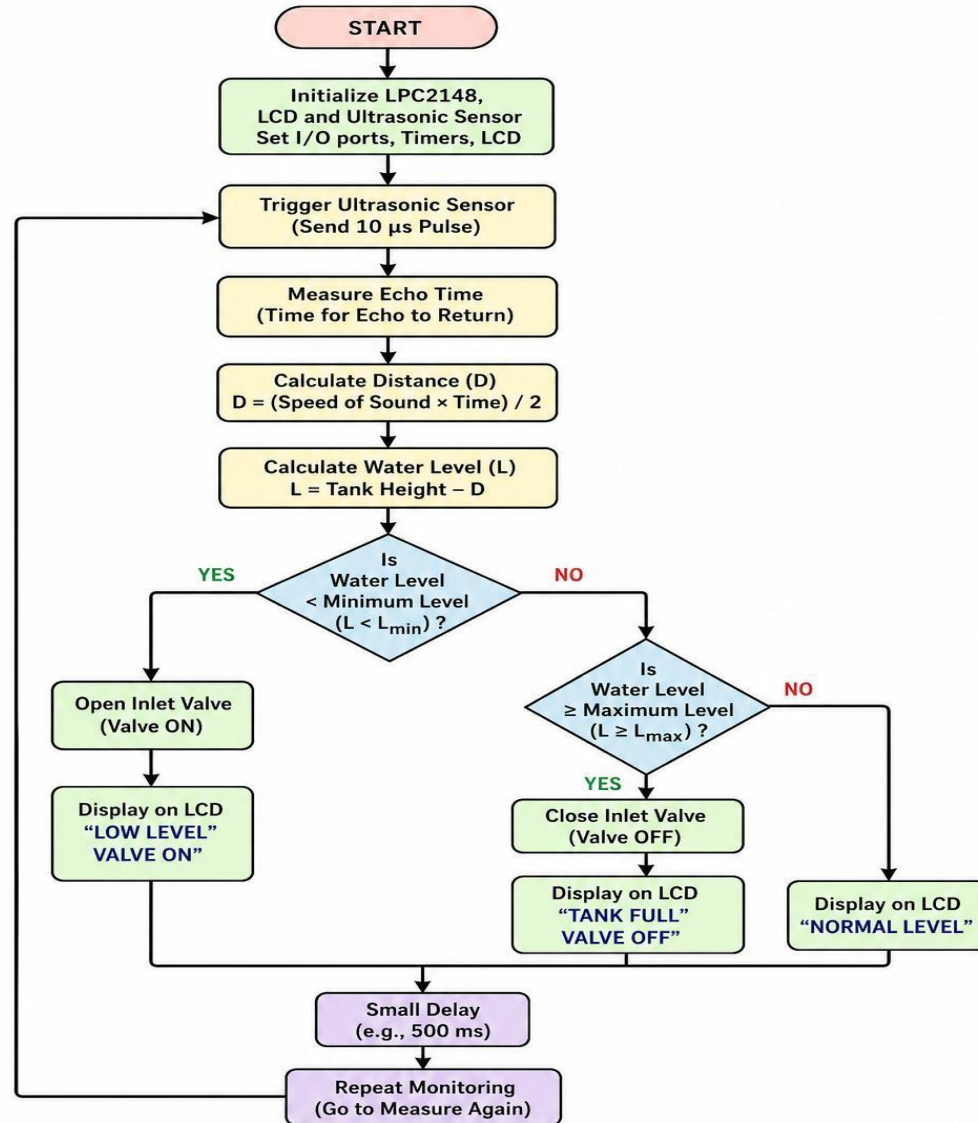


- When the water level is 100%
- The valve is off



FLOW CHART

FLOW CHART – CONTACTLESS LIQUID LEVEL CONTROLLER
USING LPC2148



RESULTS AND CONCLUSION:

Results:

1. Accurate contactless water level monitoring.
2. Automatic control of the DC Motor.
3. Prevention of tank overflow.
4. Protection against dry-running.
5. Real-time level indication on LCD.
6. Reduced manual intervention.

Conclusion:

The LPC2148-based Contactless Liquid Level Controller provides a simple and efficient solution for automatic tank water management. By combining an Ultrasonic Sensor with a DC Motor control mechanism, the system ensures reliable operation, minimizes water wastage, and enhances overall convenience in domestic and industrial applications.

THANK YOU!